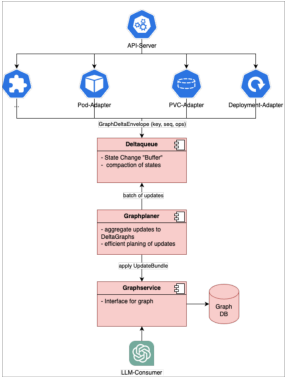
LUDWIG-MAXIMILIANS-UNIVERSITÄT MÜNCHEN

From Snapshots to Streams: A Live State Graph Approach for LLM-Based Root-Cause-Analysis in Kubernetes

OliverStempel

April 9, 2026



Outline

1. Introduction
2. Background
3. Research Question
4. Methodology
5. Evaluation
6. Contributions

Introduction

- Kubernetes, Containerisation and cloud-native technologies have become industry standard
- Key challenges for operation: perceived complexity, lack of training [3]

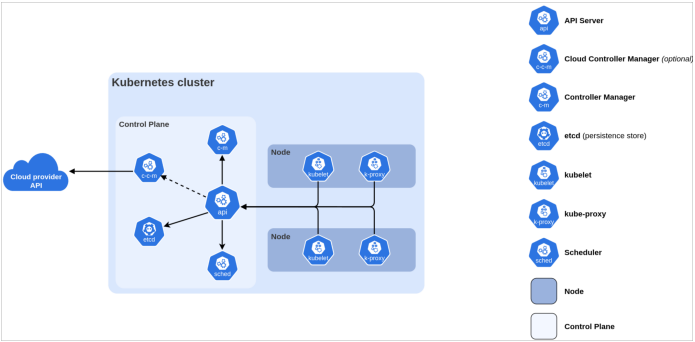
- IT-Downtime generate enormous costs for companies [8]
- Root cause analysis (RCA) as key bottleneck [5]

$\Rightarrow$ LLM-based assistance systems promise cognitive reduction and timely detection of issues

Background

Kubernetes (k8s):

- Open-source, feature-rich container orchestrator
- Deployments through declarative manifests
- Controller continuously reconcile actual with desried states



Background

Error sources:

- Unforeseen events: node dies, network issues, ...
- Syntactically correct but semantically incorrect manifests

Error handling:

- Reading docs
- Manually debugging through API-resources with the CLI
- Applying fixes

```yaml
---
apiVersion: v1
kind: ConfigMap
metadata:
  name: app-config
data:
  key: value
---
apiVersion: apps/v1
kind: Deployment
metadata:
  name: my-app
spec:
  replicas: 1
  selector:
    matchLabels:
      app: my-app
  template:
    metadata:
      labels:
        app: my-app
    spec:
      containers:
        - name: main
          image: nginx:1.27
          volumeMounts:
            - name: config
              mountPath: /etc/config
      volumes:
        - name: config
          configMap:
            name: app-config-typo
```

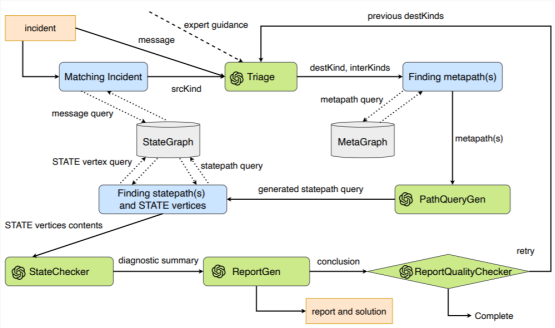
Background

Synergy-RCA [9]:

- periodically maps cluster snapshots to graph
- leverages LLMs to generate queries, checks, reports

Limitations:

- graph is not live → doesn't capture all events
- heavylifting of LLM prompts and expert knowledge



Research Question

Research Question:

"Does a live state graph for Kubernetes and predefined traversal tools improve the quality, speed and interpretability under dynamic cluster conditions compared to snapshot-based approaches."

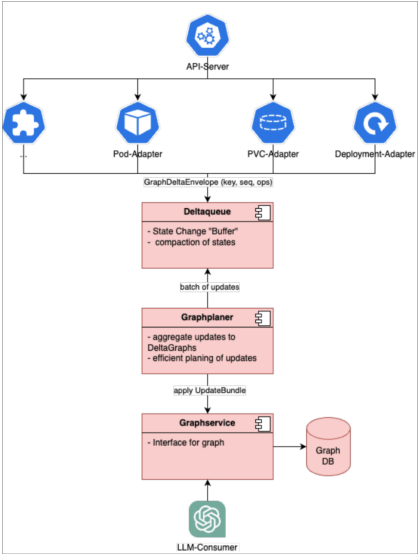
Methodology

Delta Queue :

- Buffering stream [2]
- Selection and compaction [6, 4]

Graph Planer [7]:

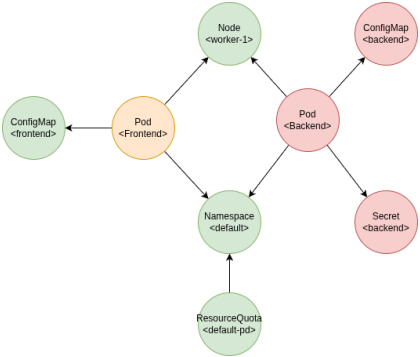
- Aggregation, ordering and efficient planning of updates



Methodology

Graph service:

- graph mine initial context
- refine context with predefined traversal tool
- trace trajectory for interpretability [1]



Evaluation

Fault injection:

- Dataset of synthetic workflows with malformed manifests
- Each workflow:
  - apply to cluster
  - run RCA
  - cleanup
- Reproducible across both systems for fair comparison
- Example: *Deployment with invalid image tag*

Ablations parameters:

- Graphplaner:
  - batch or interval based
  - batch size and interval time
- Graph traversal:
  - no graph mining / no refinement
  - different graph mining algorithms

Metrics:

- resemble Synergy-RCA metrics
  - time-to-error-detection
  - correctness
  - token-usage

Contributions

- live-state graph approach under dynamic conditions
- Introduction of LLM-friendly and interpretable traversal interface
- Design of RCA-Dataset for Kubernetes
- Derives recommendations for future AI-based RCA-Systems

References I

Sindhuja Chaduvula, Jessee Ho, Kina Kim, Aravind Narayanan, Mahshid Alinoori, Muskan Garg, Dhanesh Ramachandram, and Shaina Raza.
From features to actions: Explainability in traditional and agentic ai systems, 2026.

Raymond Cheng, Ji Hong, Aapo Kyrola, Youshan Miao, Xuetian Weng, Ming Wu, Fan Yang, Lidong Zhou, Feng Zhao, and Enhong Chen.
Kineograph: taking the pulse of a fast-changing and connected world.
In *Proceedings of the 7th ACM European Conference on Computer Systems*, EuroSys '12, page 85–98, New York, NY, USA, 2012. Association for Computing Machinery.

Cloud Native Computing Foundation.
Cloud native 2024: Approaching a decade of code, cloud, and change, 2024.
[Online].

References II

Martin Kleppmann.
Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems.
O'Reilly Media, 2017.

Komodor.
Komodor 2025 enterprise kubernetes report.
Industry report, Komodor Automation Ltd., September 2025.

Jay Kreps.
The log: What every software engineer should know about real-time data's unifying abstraction, 2013.
LinkedIn Engineering Blog.

References III

Mugilan Mariappan and Keval Vora.
Graphbolt: Dependency-driven synchronous processing of streaming graphs.
In *Proceedings of the Fourteenth EuroSys Conference 2019*, EuroSys '19, New York, NY, USA, 2019. Association for Computing Machinery.

Splunk.
The hidden costs of downtime: The $400b problem facing the global 2000.
Industry report, Splunk Inc., 2024.
In partnership with Oxford Economics.

Yong Xiang, Charley Peter Chen, Liyi Zeng, Wei Yin, Xin Liu, Hu Li, and Wei Xu.
Simplifying root cause analysis in kubernetes with stategraph and llm, 2025.